

Asymptotically Exact Orthogonal Cutdowns for Compact C^* -Algebras

Candidate partial answer to arXiv:math/0003152

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Status: candidate partial result, likely valid. The second source question is answered for compact (dual, or annihilator) C^* -algebras. The general case and the first source question remain open here.

1 Source question

Let A be a C^* -algebra and let $(\phi_m) \subset A^*$ be a bounded sequence spanning ℓ_1 almost isometrically. Proposition 1.3 of Pfitzner [1] says that, for every fixed $\varepsilon > 0$, one can find a subsequence (ϕ_{m_n}) and pairwise orthogonal positive normalized elements (a_n) and (b_n) in A such that

$$\|\phi_{m_n} - b_n \phi_{m_n} a_n\| < \varepsilon \quad (n \in \mathbb{N}).$$

The second question in Section 6 asks whether these objects can instead be chosen so that

$$\|\phi_{m_n} - b_n \phi_{m_n} a_n\| \longrightarrow 0. \tag{1}$$

Here $(b\phi a)(x) = \phi(axb)$. The theorem below proves this for every compact C^* -algebra, meaning a c_0 -direct sum of elementary compact operator algebras.

2 Finite-rank cutdowns of orthogonal functionals

We use the standard description

$$A = \bigoplus_{i \in I}^{c_0} \mathcal{K}(H_i), \quad A^* = \bigoplus_{i \in I}^{\ell_1} S_1(H_i), \quad A^{**} = \bigoplus_{i \in I}^{\ell_\infty} B(H_i).$$

A functional $\psi \in A^*$ is represented by a trace-class family $T = (T_i)$, with $\psi(x) = \sum_i \text{Tr}(T_i x_i)$ and $\|\psi\| = \sum_i \|T_i\|_1$.

Lemma 1. *Let (ψ_n) be pairwise orthogonal functionals in A^* , and let (η_n) be positive. For every n with $\psi_n \neq 0$ there are finite-rank projections $e_n, f_n \in A$ such that both families are pairwise orthogonal and*

$$\|\psi_n - f_n \psi_n e_n\| < \eta_n.$$

Proof. Write $T^{(n)} = (T_i^{(n)})$ for the trace-class family representing ψ_n . Choose a finite set of coordinates and, in each chosen coordinate, finitely many singular vectors so that the corresponding finite singular-value truncation $S^{(n)}$ satisfies $\|T^{(n)} - S^{(n)}\|_1 < \eta_n$. If e_n and f_n are respectively the

right and left support projections of $S^{(n)}$, then they have finite support in I , are finite-rank in every coordinate, and hence belong to A . They satisfy

$$f_n T^{(n)} e_n = S^{(n)},$$

after taking the truncation in a singular-value decomposition.

Orthogonality of two normal functionals is equivalent to orthogonality of their right support projections and of their left support projections in A^{**} . Since e_n and f_n are subprojections of those supports, the e_n are pairwise orthogonal, as are the f_n . Finally, the functional $f_n \psi_n e_n$ is represented by $f_n T^{(n)} e_n = S^{(n)}$, so the asserted norm estimate is exactly the trace-norm estimate above. $\square$

3 The partial answer

Theorem 2. *Let $A = \bigoplus_{i \in I}^{\text{co}} \mathcal{K}(H_i)$ be a compact C^* -algebra, and let $(\phi_m) \subset A^*$ be a bounded sequence spanning ℓ_1 almost isometrically. Then there are a subsequence (ϕ_{m_n}) and pairwise orthogonal positive normalized elements (a_n) and (b_n) in A such that*

$$\|\phi_{m_n} - b_n \phi_{m_n} a_n\| \longrightarrow 0.$$

The elements a_n, b_n may be chosen to be finite-rank projections.

Proof. The von Neumann algebra A^{**} has predual canonically equal to A^* . Pfizner's orthogonalization theorem [1, Theorem 1.2] (equivalently, the JBW*-triple extension [2, Theorem 4.1]) yields a subsequence, still denoted (ϕ_{m_n}) , and pairwise orthogonal functionals $\psi_n \in A^*$ such that

$$\|\phi_{m_n} - \psi_n\| \longrightarrow 0. \quad (2)$$

First suppose, after discarding finitely many terms, that $\psi_n \neq 0$. Apply Lemma 1 with $\eta_n \downarrow 0$, and put $a_n = e_n$, $b_n = f_n$. Left and right multiplication by contractions are contractive on A^* , whence

$$\begin{aligned} \|\phi_{m_n} - b_n \phi_{m_n} a_n\| &\leq \|\phi_{m_n} - \psi_n\| + \|\psi_n - b_n \psi_n a_n\| + \|b_n(\psi_n - \phi_{m_n})a_n\| \\ &< 2\|\phi_{m_n} - \psi_n\| + \eta_n \longrightarrow 0. \end{aligned}$$

It remains only to handle the possible zero terms. If infinitely many ψ_n vanish, then along that subsequence (2) gives $\|\phi_{m_n}\| \rightarrow 0$. The normalized sequence $(\phi_m/\|\phi_m\|)$ spans ℓ_1 isomorphically, so A^* , and hence A , is infinite-dimensional. An infinite-dimensional compact C^* -algebra contains an infinite sequence of pairwise orthogonal finite-rank projections; choose two such sequences (a_n) and (b_n) . Then

$$\|\phi_{m_n} - b_n \phi_{m_n} a_n\| \leq 2\|\phi_{m_n}\| \longrightarrow 0.$$

This completes both cases. $\square$

4 Why this does not settle the general case

The proof first orthogonalizes in the von Neumann envelope A^{**} and then cuts the orthogonal support projections down inside A . In a compact C^* -algebra, finite singular-value truncation supplies these inner cutdowns while preserving pairwise orthogonality. For a general C^* -algebra the support projections remain in A^{**} , and the missing step is a countable separation theorem producing pairwise orthogonal elements of A with individual support errors tending to zero.

A direct attempt to let the distortion parameter tend to zero in the proof of Proposition 1.3 encounters the same issue in quantitative form: leakage into all previously chosen closed supports accumulates to a fixed loss. A summable leakage schedule makes that loss small but does not make it vanish. Thus the compact-operator truncation above is a genuine additional input, not merely a reparameterization of the fixed- ε proof.

5 Literature scope

The exact title, question phrase, and citation trail were searched. The later paper [2] proves the normal orthogonal-functional theorem for JBW^* -triples and a fixed-accuracy non-normal theorem, but it does not state (1) for arbitrary C^* -algebras. No source stating the compact-algebra theorem above was located in the bounded search. Novelty confidence is therefore limited rather than definitive.

References

- [1] H. Pfitzner, *Perturbation of ℓ_1 -copies and measure convergence in preduals of von Neumann algebras*, J. Operator Theory **47** (2002), 145–167; arXiv:math/0003152.
- [2] A. M. Peralta and H. Pfitzner, *Perturbation of ℓ_1 -copies in preduals of JBW^* -triples*, J. Math. Anal. Appl. **434** (2016), 149–170; arXiv:1405.5414.